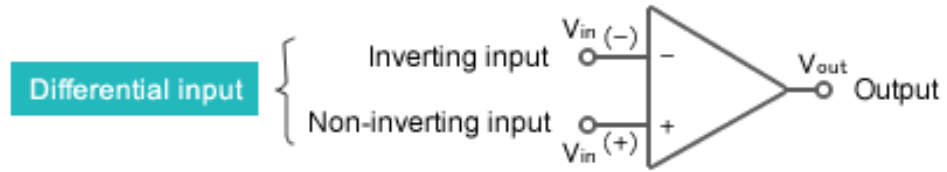


Operational Amplifiers

OPERATIONAL AMPLIFIERS (OP AMP)

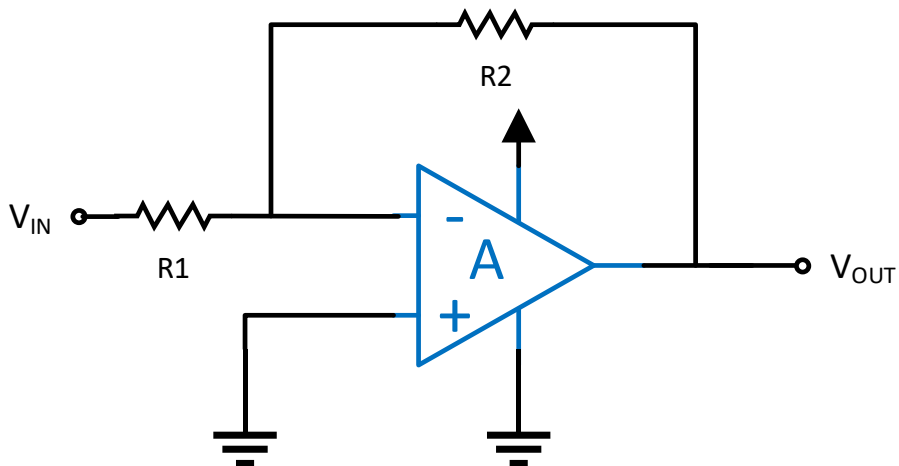
An op-amp operates on analog input. It can be used to amplify or attenuate this input, and to carry out mathematical operations such as addition, subtraction, integration, and differentiation. Because of their wide range of uses, op-amps are encountered in most electric circuits.



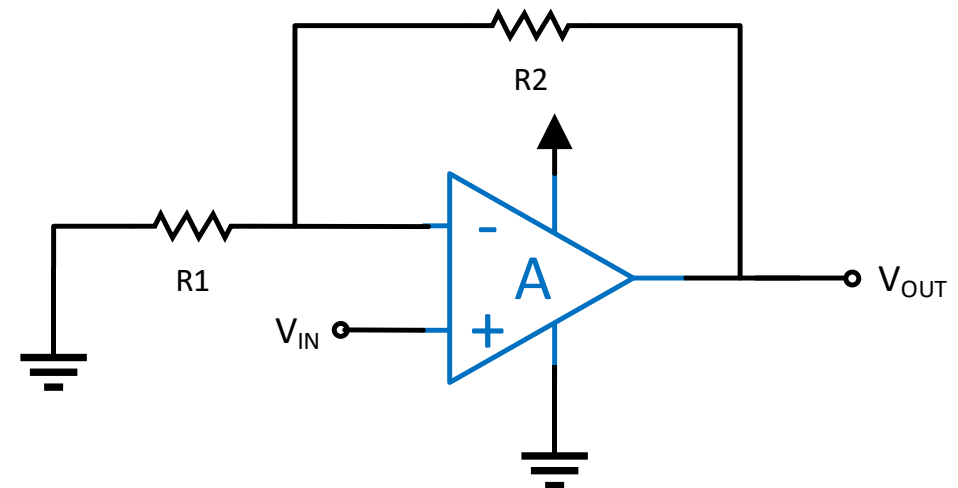
$$V_{out} = A \cdot (V_{in+} - V_{in-})$$

A: Open Loop Gain

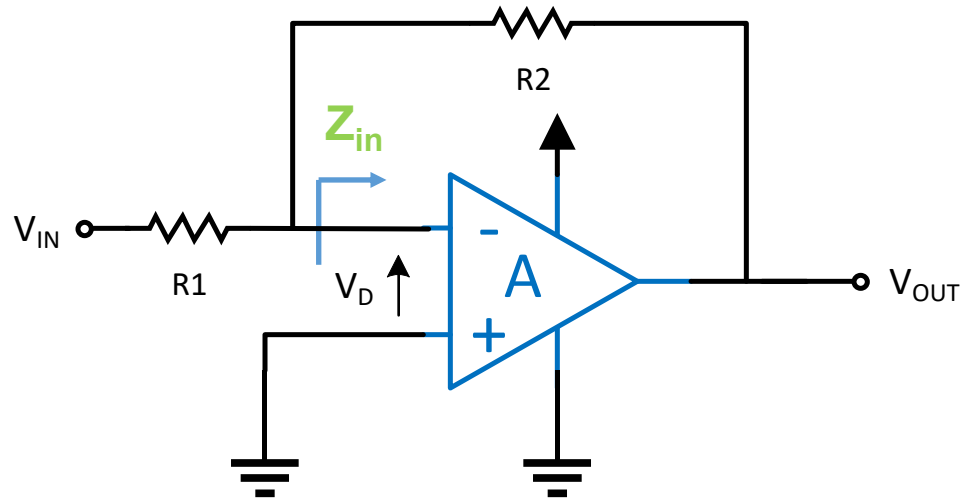
Inverting Amplifier



Non-Inverting Amplifier



INVERTING OP AMP



$$V_D + V_{R_2} + V_{out} = 0 \longrightarrow V_{R_2} = -V_{out} - V_D$$

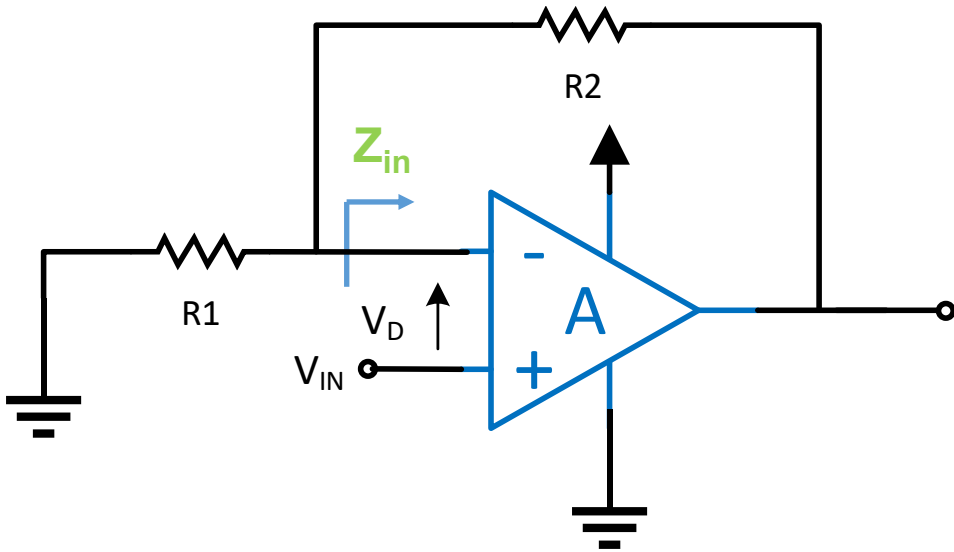
$$V_{out} = A \cdot V_D \longrightarrow V_D = \frac{V_{out}}{A}$$

$$\frac{V_{in} - V_D}{R_1} = \frac{-V_D - V_{out}}{R_2}$$

$$Gain = \frac{V_{OUT}}{V_{IN}} = \frac{-R_2 A}{R_1 + R_2 + R_1 A} = \frac{-R_2 A}{R_1 (1 + A) + R_2}$$

If $A \rightarrow \infty$, then $Z_{in} \rightarrow \infty$ and $Gain = -\frac{R_2}{R_1}$

NON-INVERTING OP AMP



$$V_{out} = (V_{IN} - V_D) + V_{R_2} \longrightarrow V_{R_2} = V_{out} - (V_{IN} - V_D)$$

$$V_{out} = A \cdot V_D \longrightarrow V_D = \frac{V_{out}}{A}$$

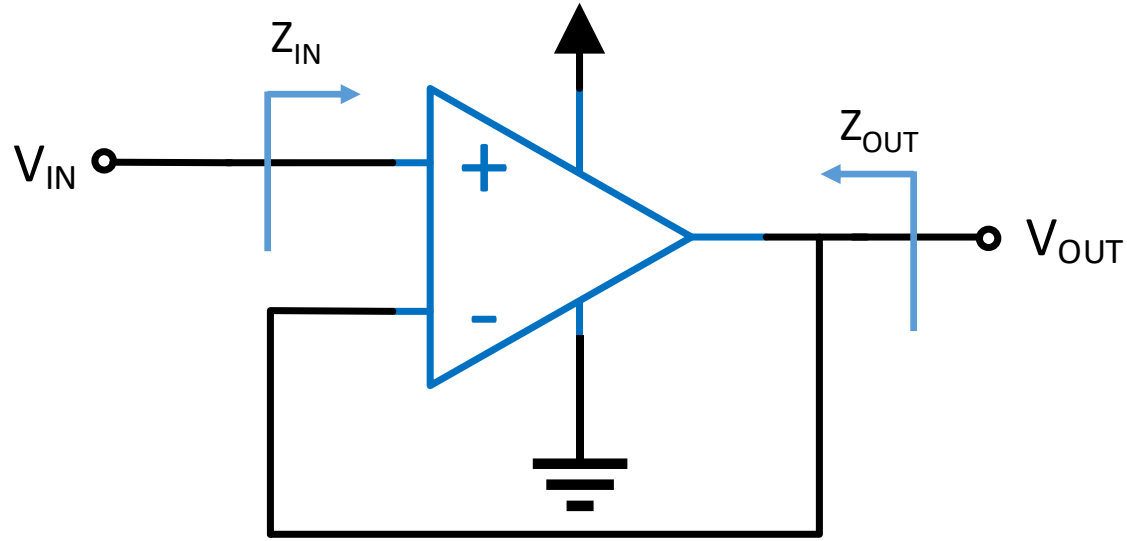
$$\frac{V_{in} - V_D}{R_1} = \frac{V_D + V_{out} - V_{IN}}{R_2}$$

$$Gain = \frac{V_{OUT}}{V_{IN}} = \frac{A(R_1 + R_2)}{R_1(1 + A) + R_2}$$

If $A \rightarrow \infty$, then $Z_{in} \rightarrow \infty$ and

$$Gain = \frac{R_1 + R_2}{R_1} = 1 + \frac{R_2}{R_1}$$

VOLTAGE FOLLOWER

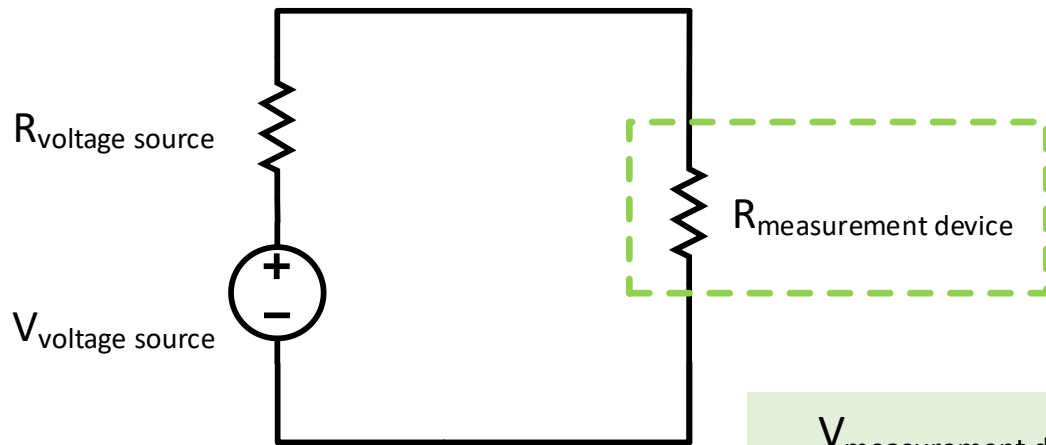


A voltage follower, also known as unity gain amplifier has a voltage gain of 1.

$$\text{Gain} = V_{\text{OUT}} / V_{\text{IN}} = 1$$

A voltage follower has high input impedance Z_{IN} and low output impedance Z_{OUT}

Why do we need a voltage follower? The following example will explain where a voltage follower can be used



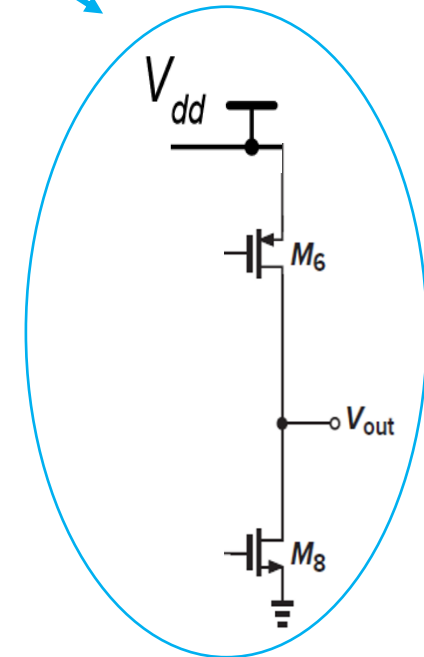
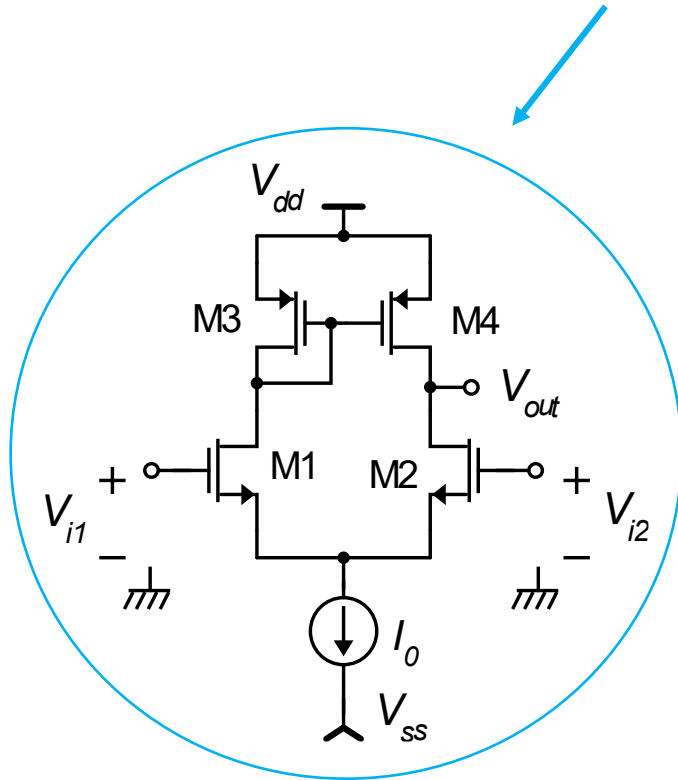
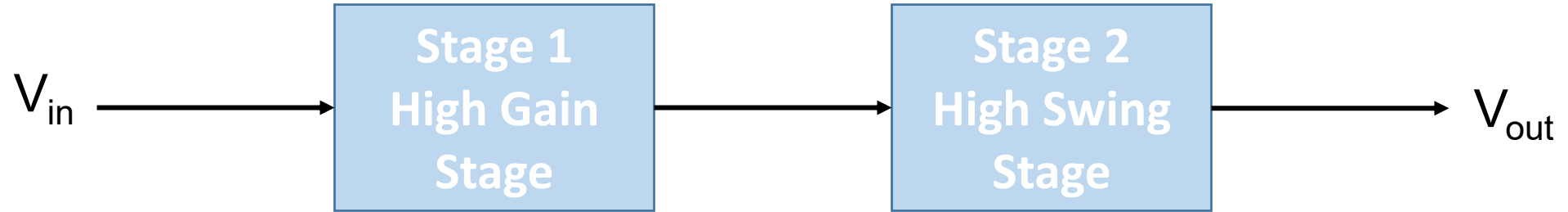
Measurement device can be an oscilloscope or a multimeter

A voltage source is not ideal has an internal resistance (and inductance)

$$V_{\text{measurement device}} = V_{\text{voltage source}} \left(\frac{R_{\text{measurement device}}}{R_{\text{measurement device}} + R_{\text{voltage source}}} \right)$$

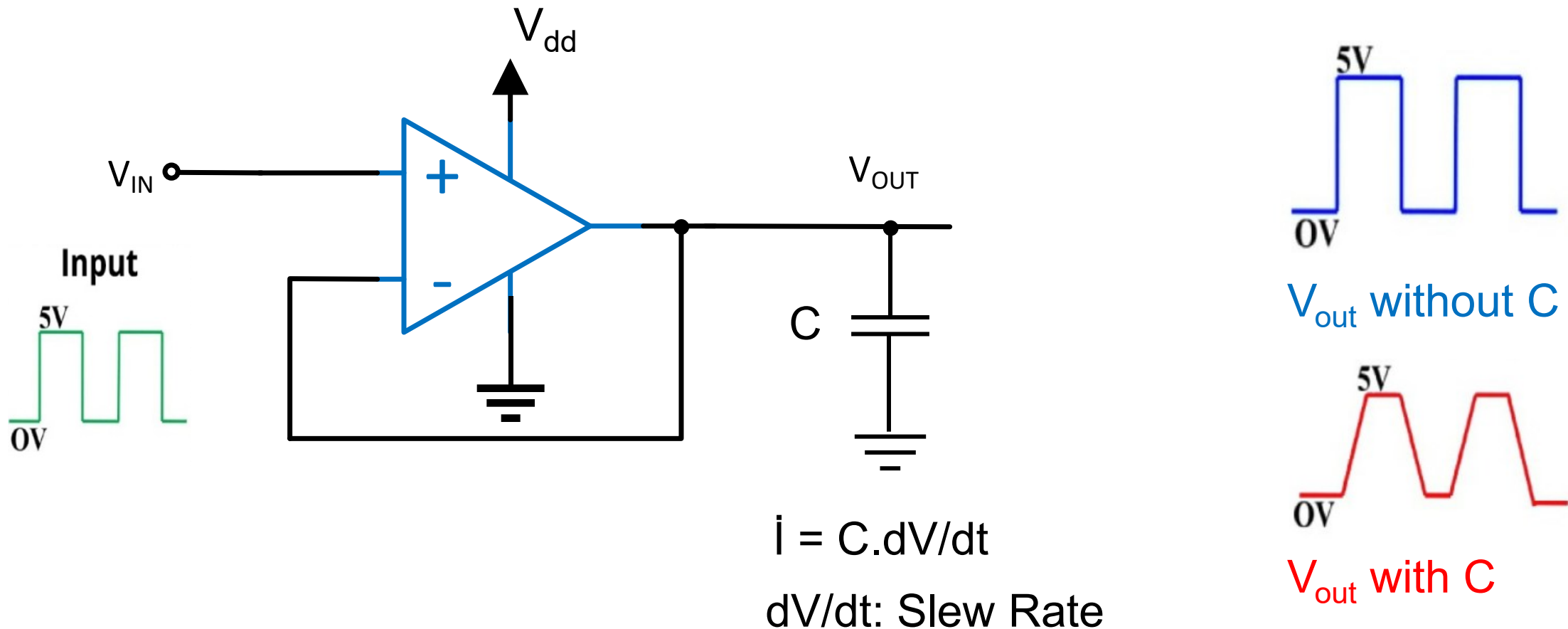
How does the design of an operational amplifier look like?

OP AMP DESIGN

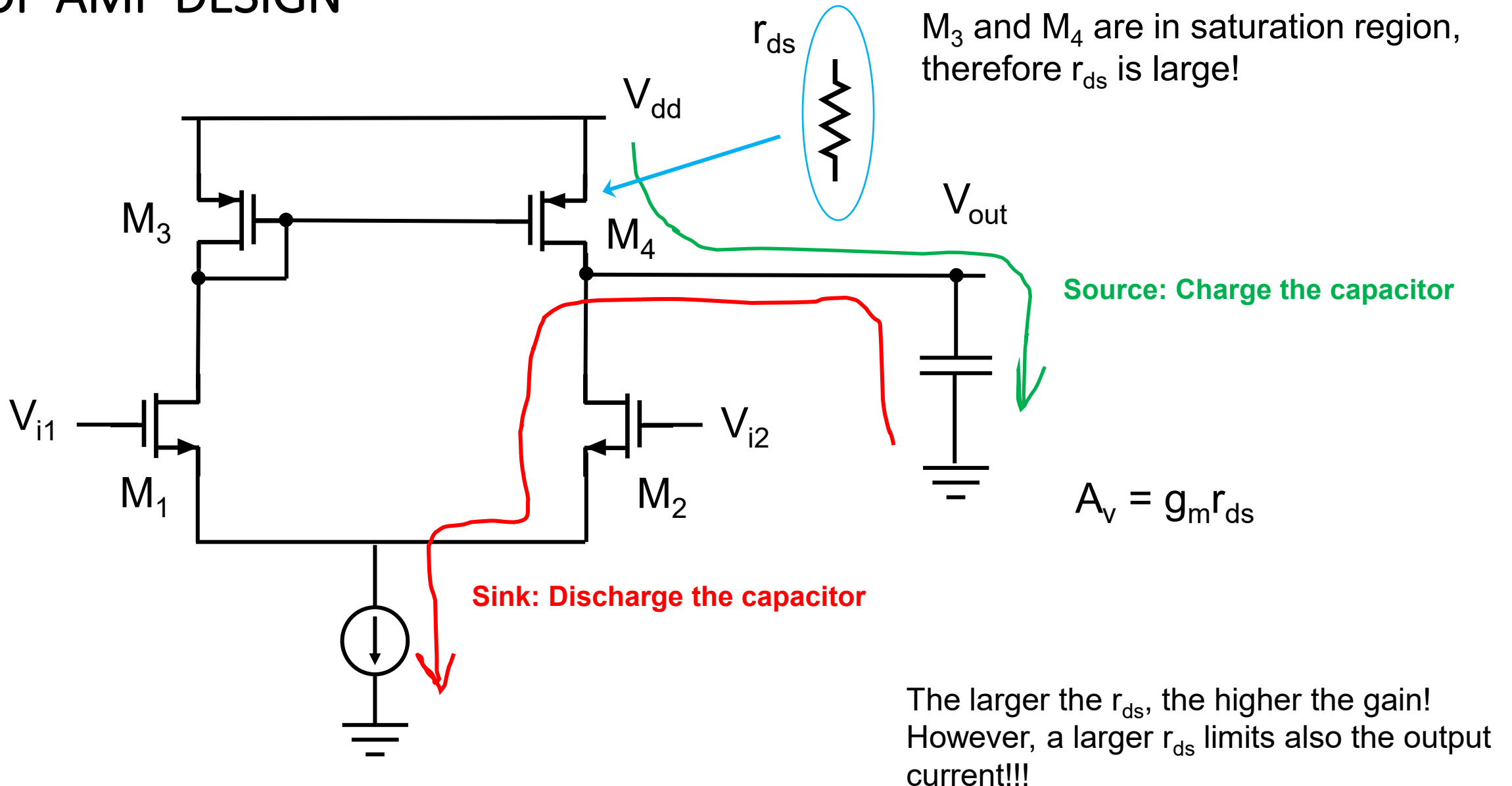


OP AMP DESIGN

The swing stage is necessary to drive capacitive and/or resistive loads without distorting the signal. The first high gain stage has a high voltage gain but this doesn't necessarily mean that the first high gain stage can also deliver large currents.



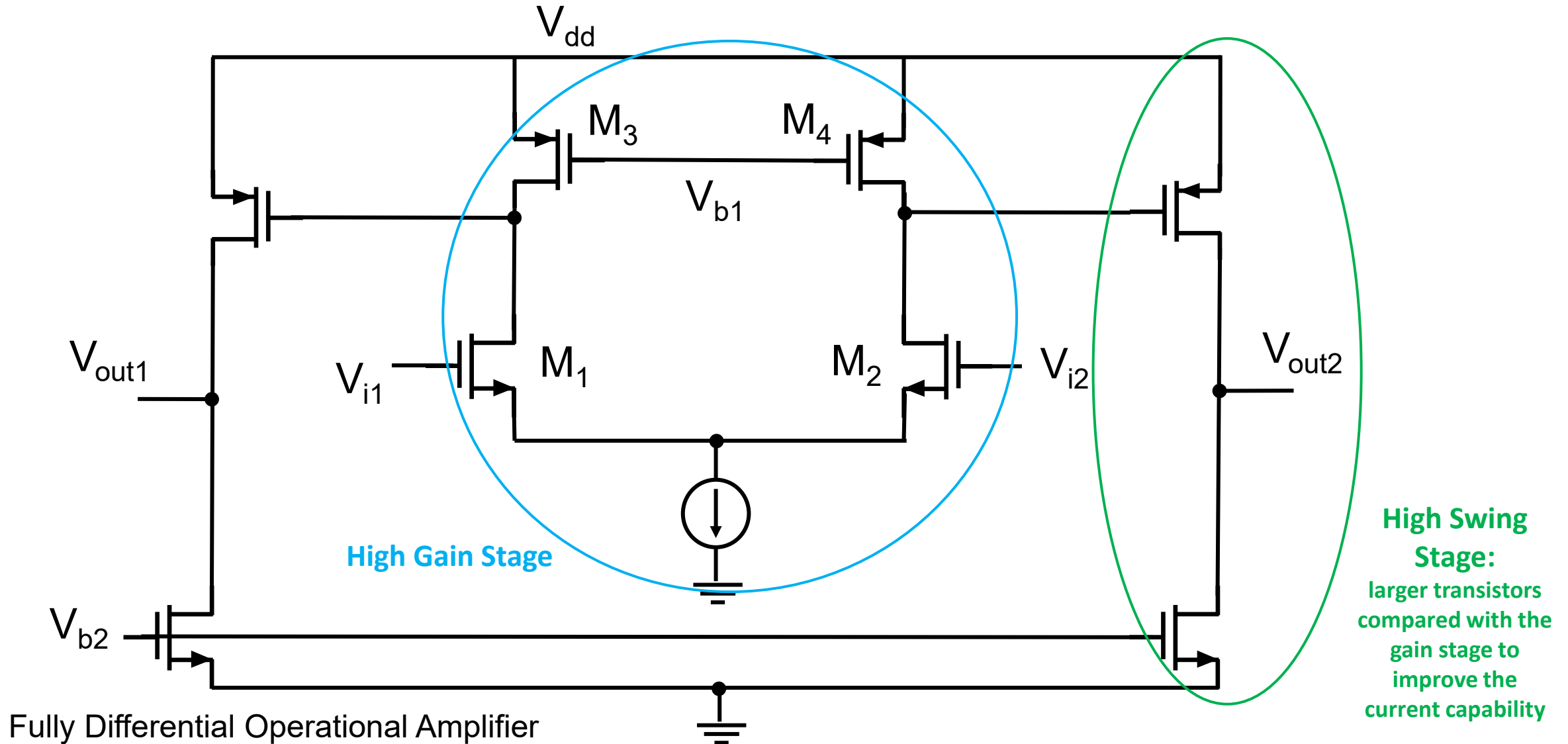
OP AMP DESIGN



Differential Amplifier with Single-Ended Output

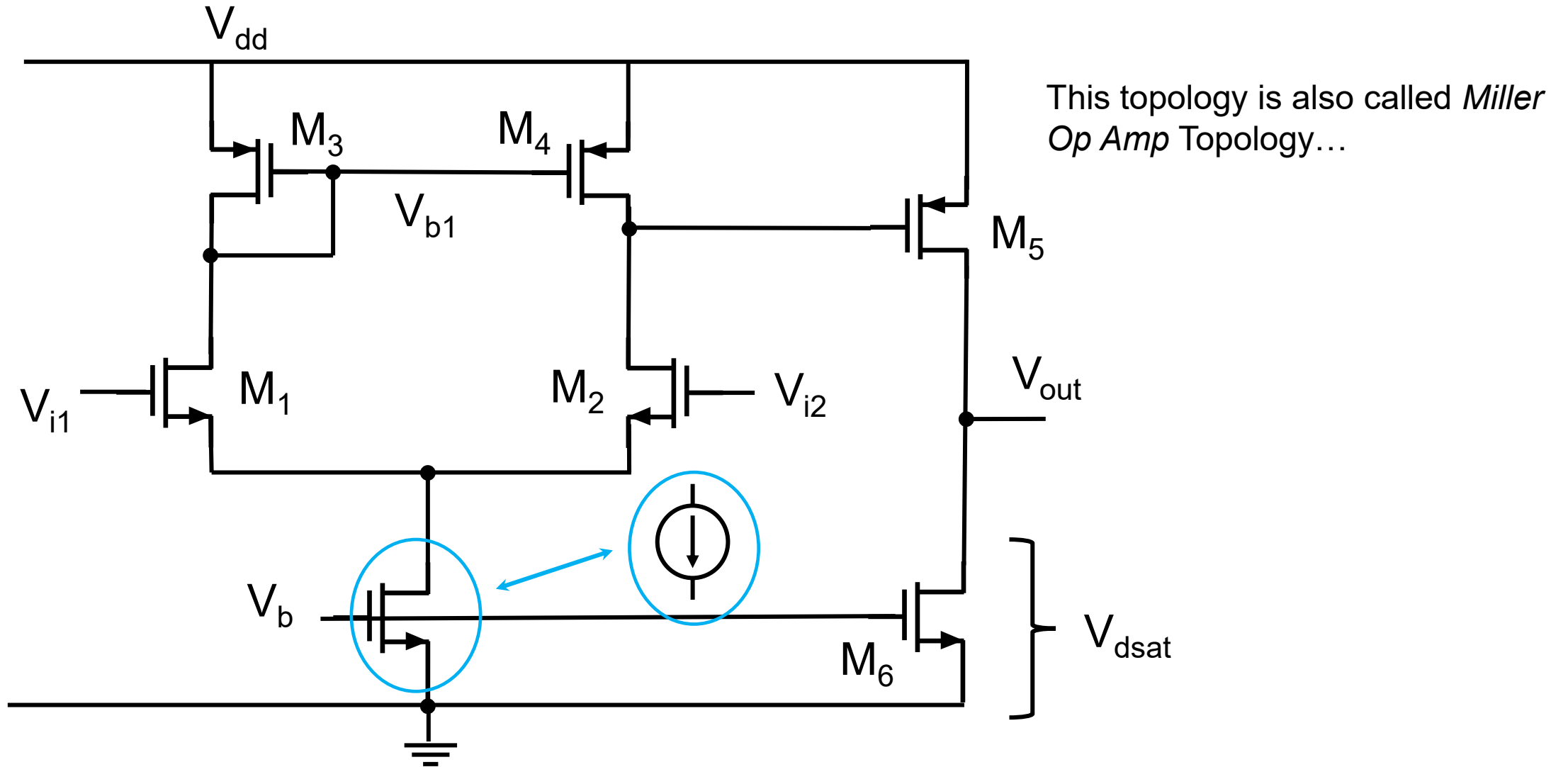
OP AMP DESIGN

The high swing stage does not need to have a high voltage gain, it can be even unity. However, it must be able to deliver high output currents!!!



OP AMP DESIGN

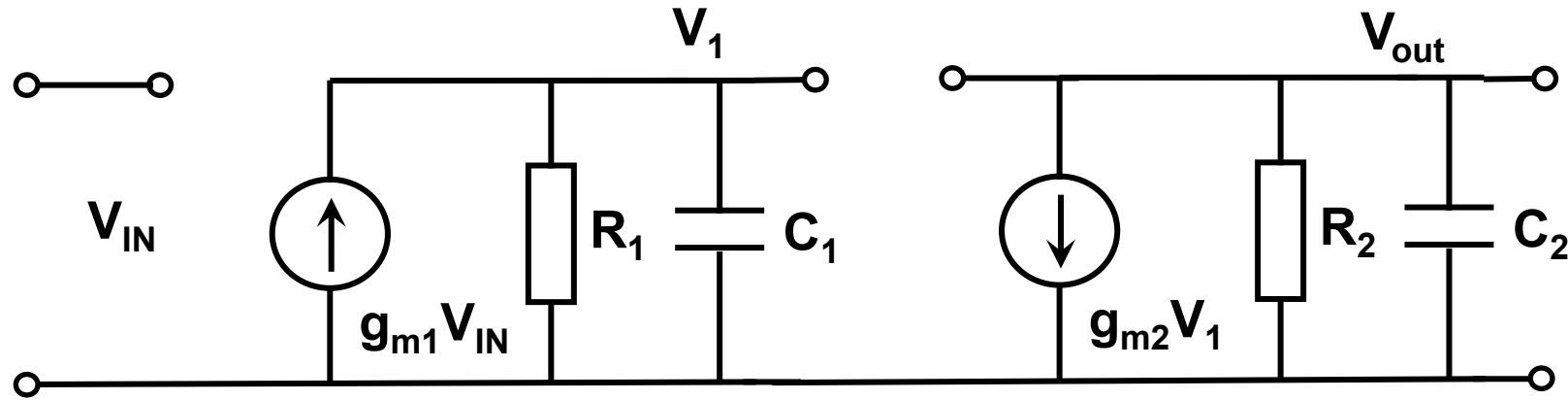
Which one of the two inputs is (+)? V_{i1} or V_{i2} ?
 V_{i2} is positive!



Operational Amplifier with single-ended output

How is V_b generated?

OP AMP DESIGN - MILLER COMPENSATION



1. Stage:

$$A_1 = g_{m1}R_1 \quad \omega_1 = \frac{1}{R_1C_1} \quad GBWP = \frac{g_{m1}}{C_1}$$

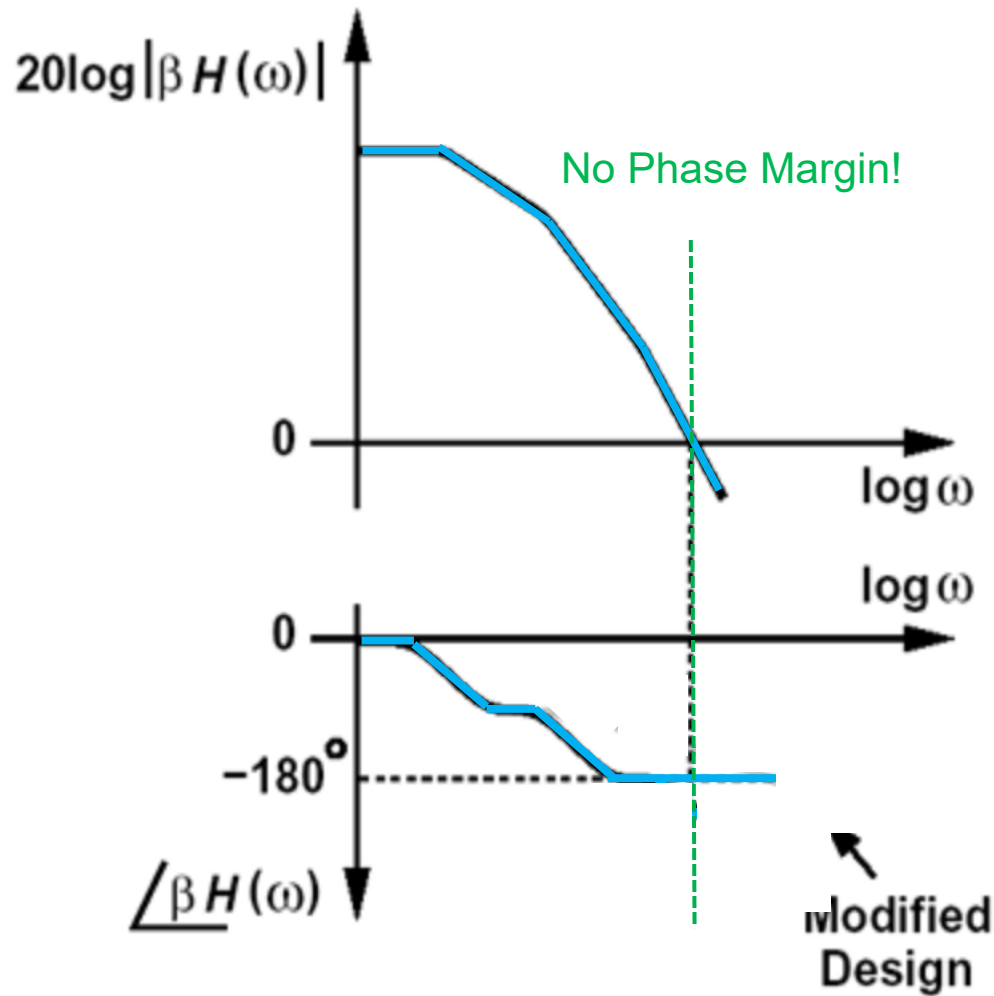
2. Stage:

$$A_2 = -g_{m2}R_2 \quad \omega_2 = \frac{1}{R_2C_2} \quad GBWP = \frac{g_{m2}}{C_2}$$

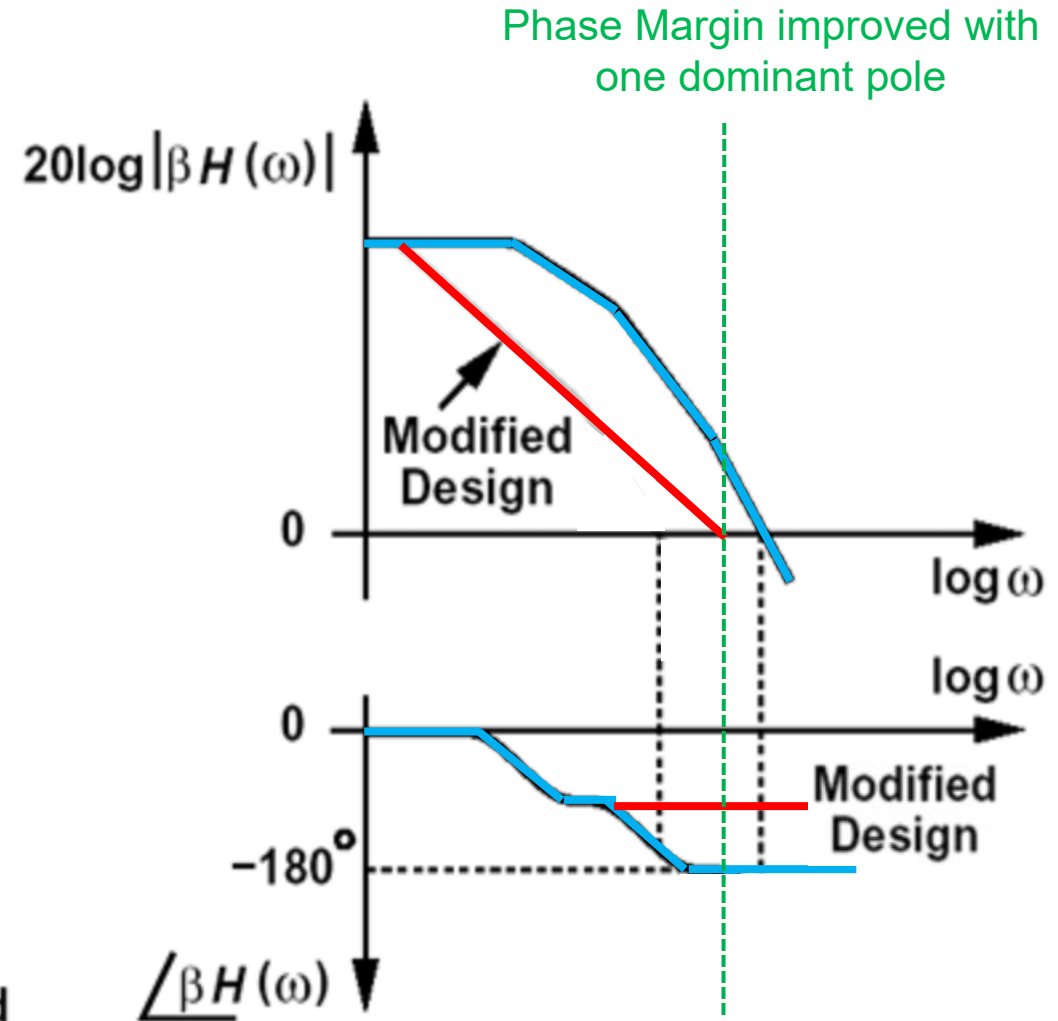
Overall:

$$A = A_1A_2 = -g_{m1}R_1g_{m2}R_2$$

OP AMP DESIGN - MILLER COMPENSATION



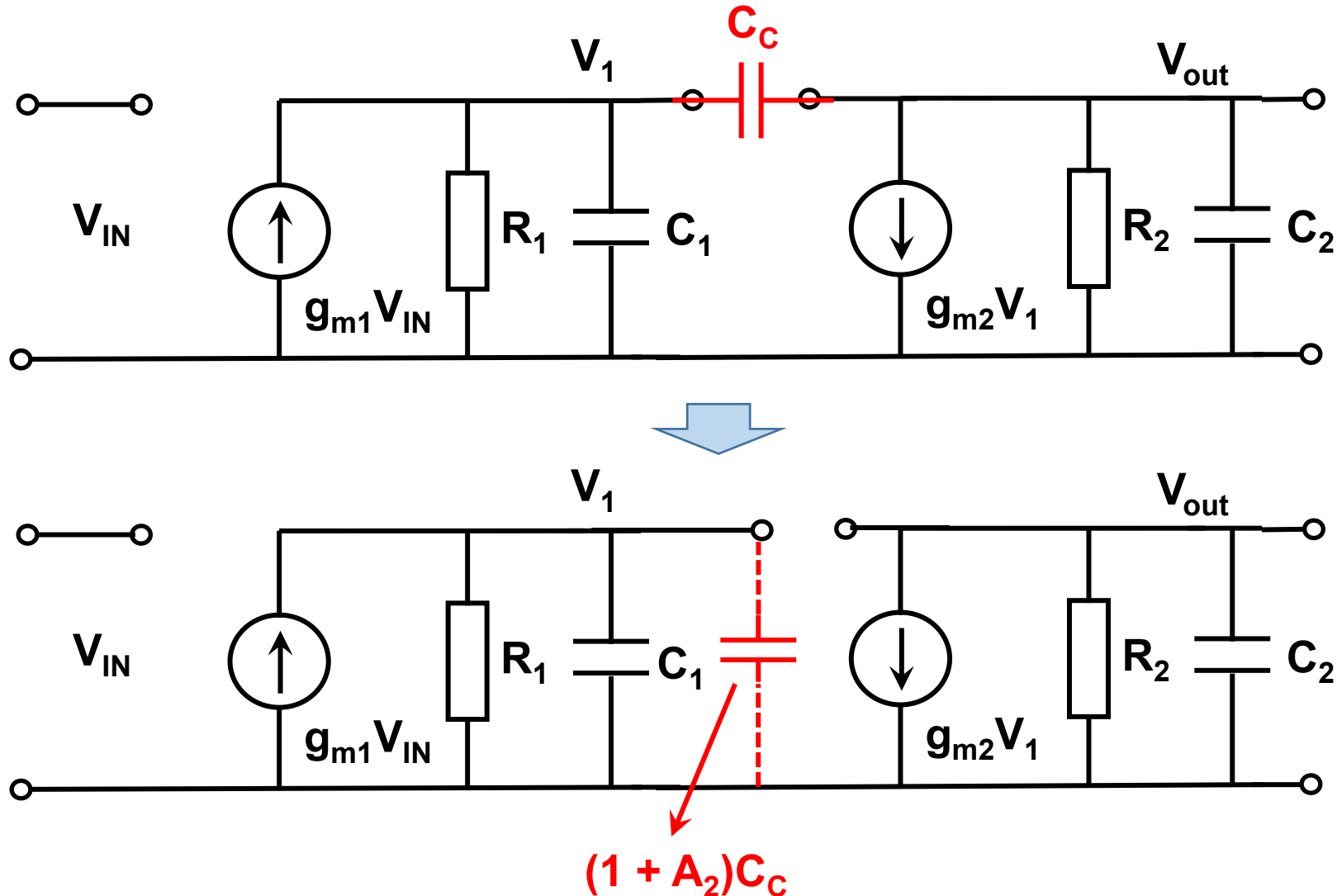
Multiple Poles lead to insufficient Phase Margin!



One Dominant Pole makes the System more stable!!!

OP AMP DESIGN - MILLER COMPENSATION

We add a Miller Capacitance between the input and the output of the second stage



*Please see the slides
«Frequency Response of
Amplifiers for the derivations»*

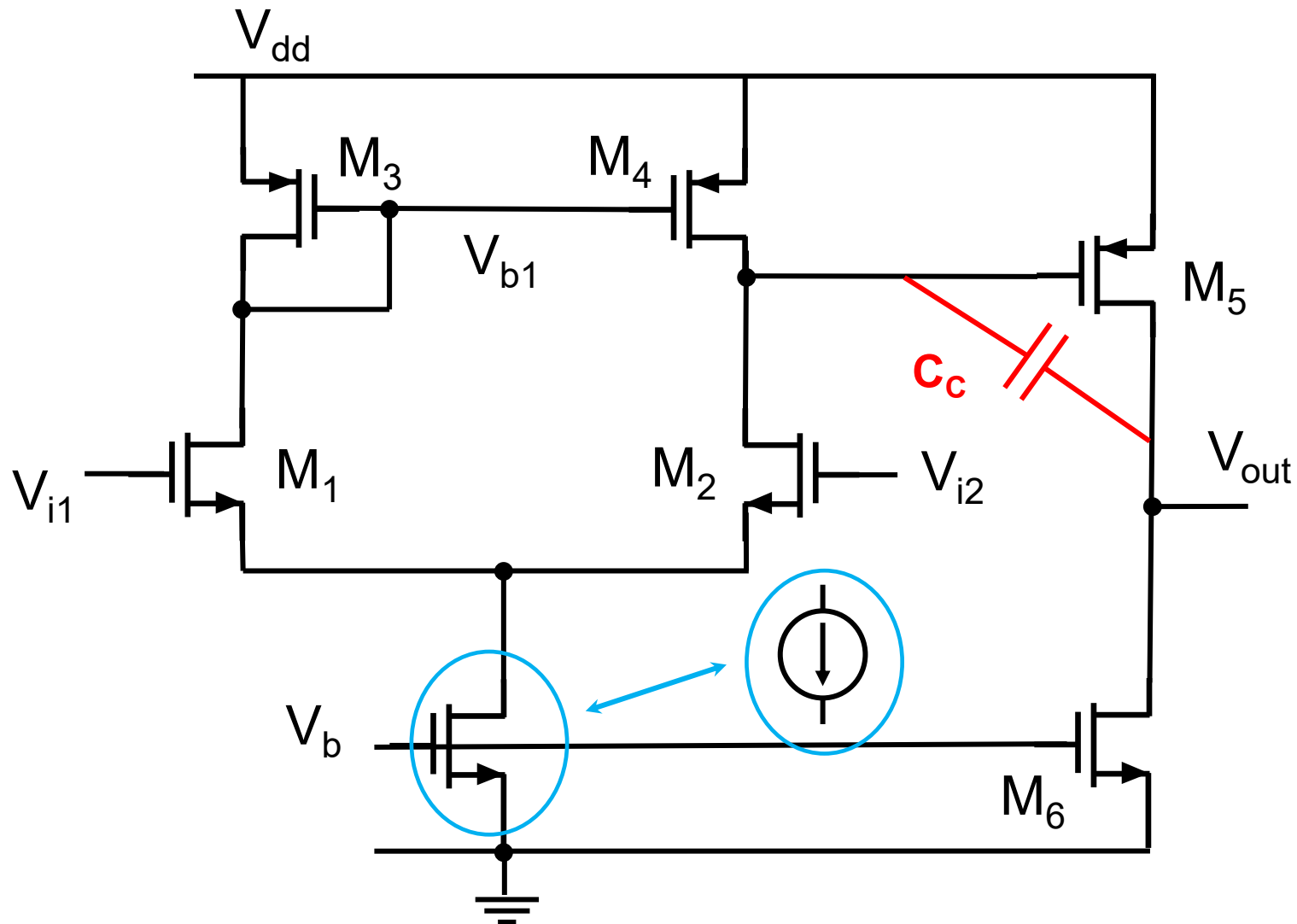
$$\begin{aligned} C_1' &= C_1 + (1 + A_2)C_C \\ &= C_1 + (1 + g_{m2}R_2)C_C \\ &= C_1 + C_C + g_{m2}R_2C_C \end{aligned}$$

The new pole can be calculated as:

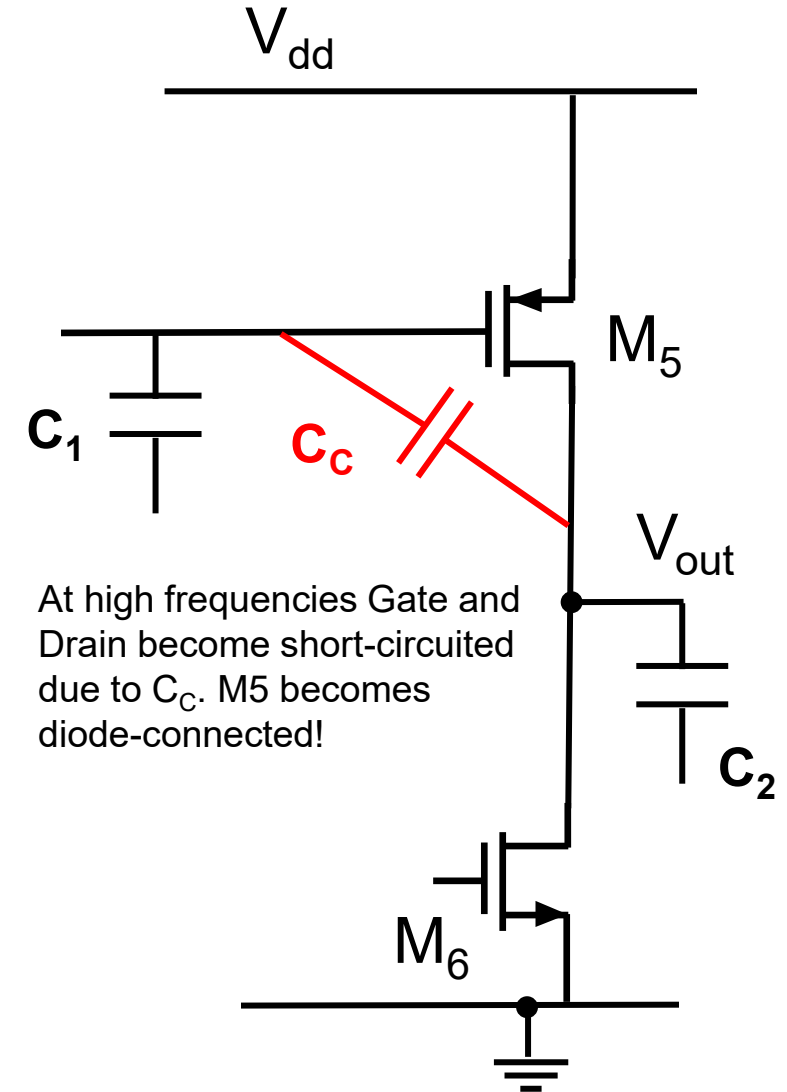
$$\omega'_{p1} = \frac{1}{R_1 C_1'} \approx \frac{1}{g_{m2} R_2 R_1 C_C}$$

$$\omega'_{p1} \ll \omega_{p1}$$

OP AMP DESIGN - MILLER COMPENSATION



C_c causes also a zero but this will be neglected within the scope of this course.



OP AMP DESIGN - MILLER COMPENSATION

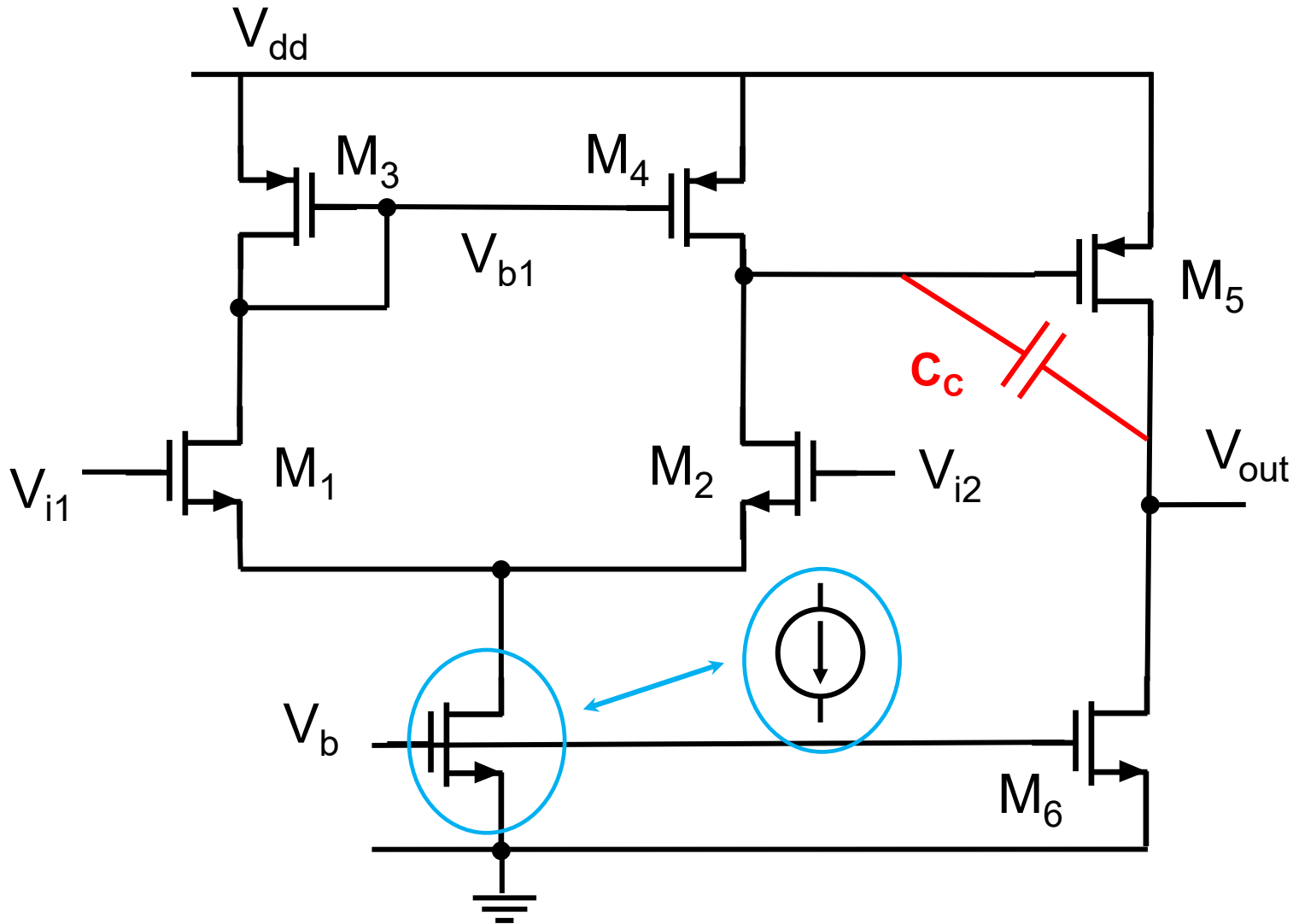
A diode-connected transistor has an output resistance of $1/g_{m2}$, thus the second pole can be written as:

$$\omega'_{p2} = \frac{g_{m2}}{C_1 + C_2} = \frac{1}{(1/g_{m2}) \cdot (C_1 + C_2)}$$

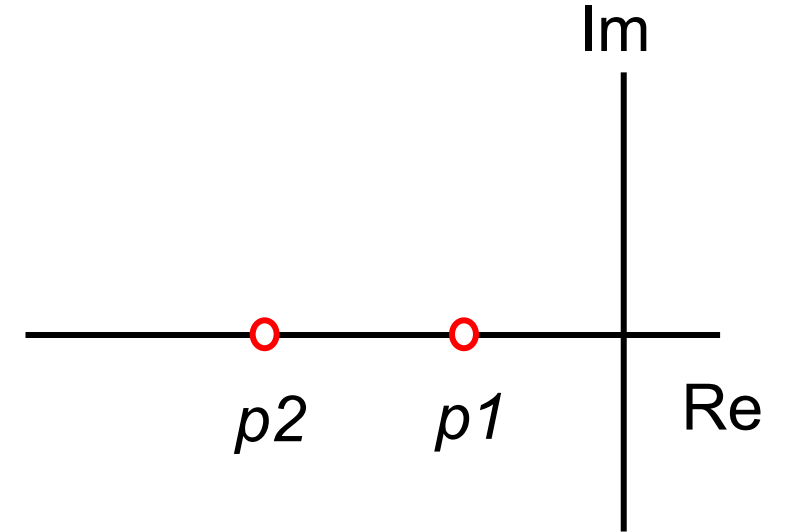
$$\omega_{p2} = \frac{1}{R_2 C_2}$$

$1/g_{m2}$ few $k\Omega$, R_2 hundreds of $k\Omega \rightarrow 1/g_{m2} < R_2$, therefore $\omega'_{p2} > \omega_{p2}$

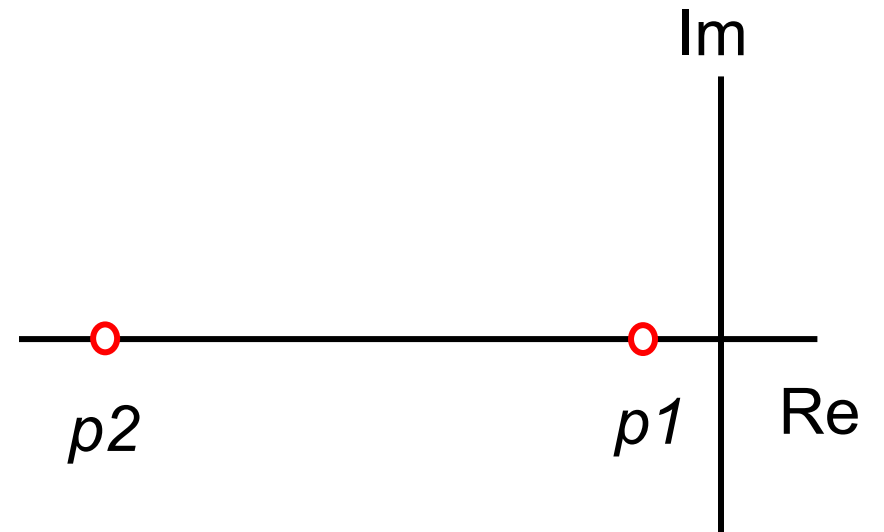
OP AMP DESIGN - MILLER COMPENSATION



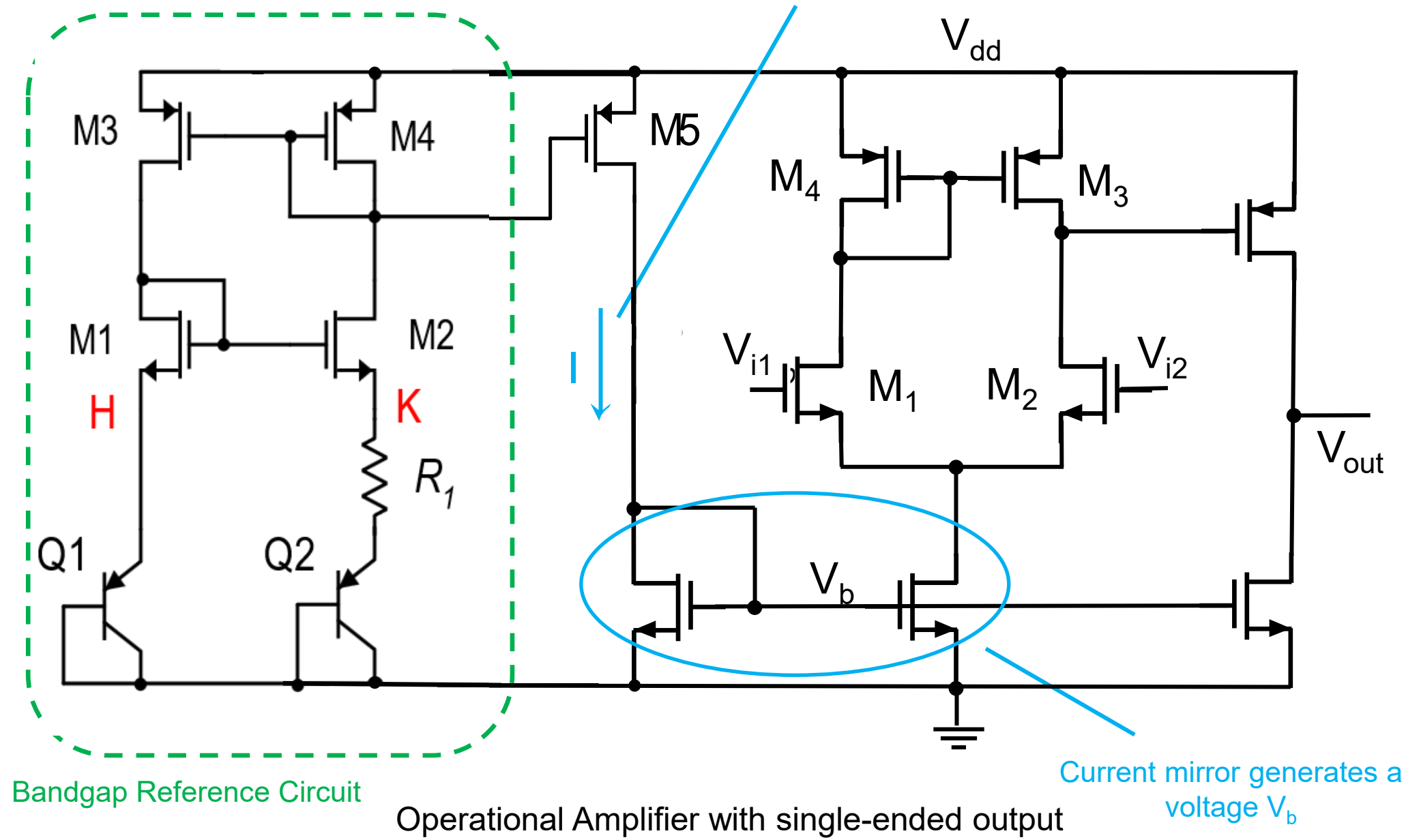
Poles without Miller Compensation



Poles with Miller Compensation

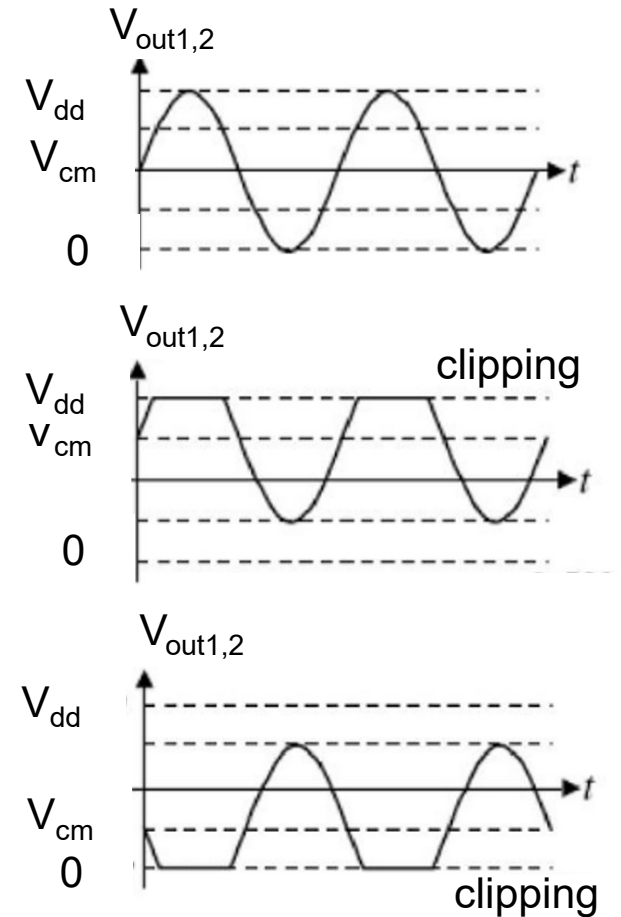
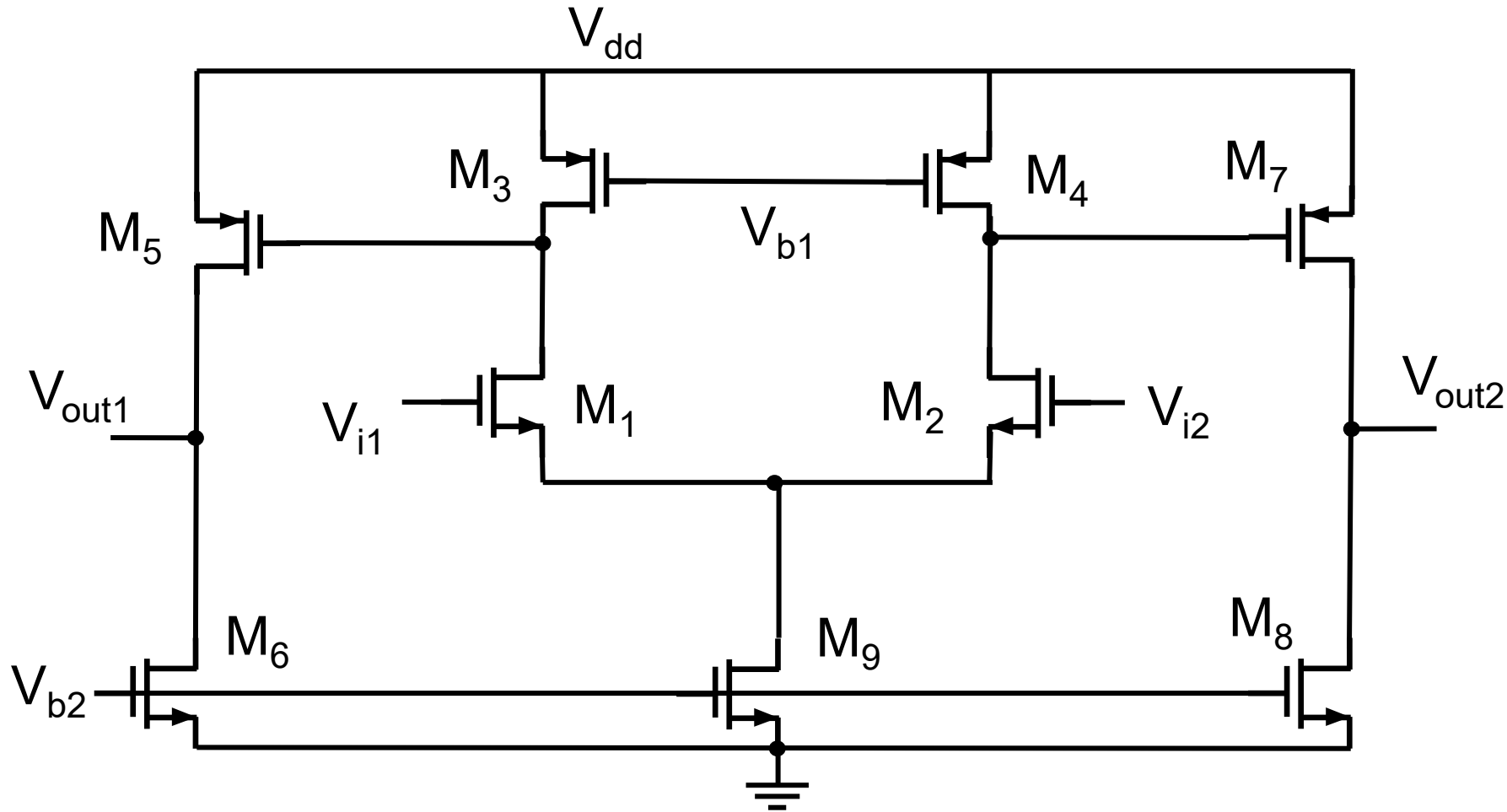


OP AMP DESIGN



OP AMP DESIGN - FULLY DIFFERENTIAL OPERATIONAL AMPLIFIER

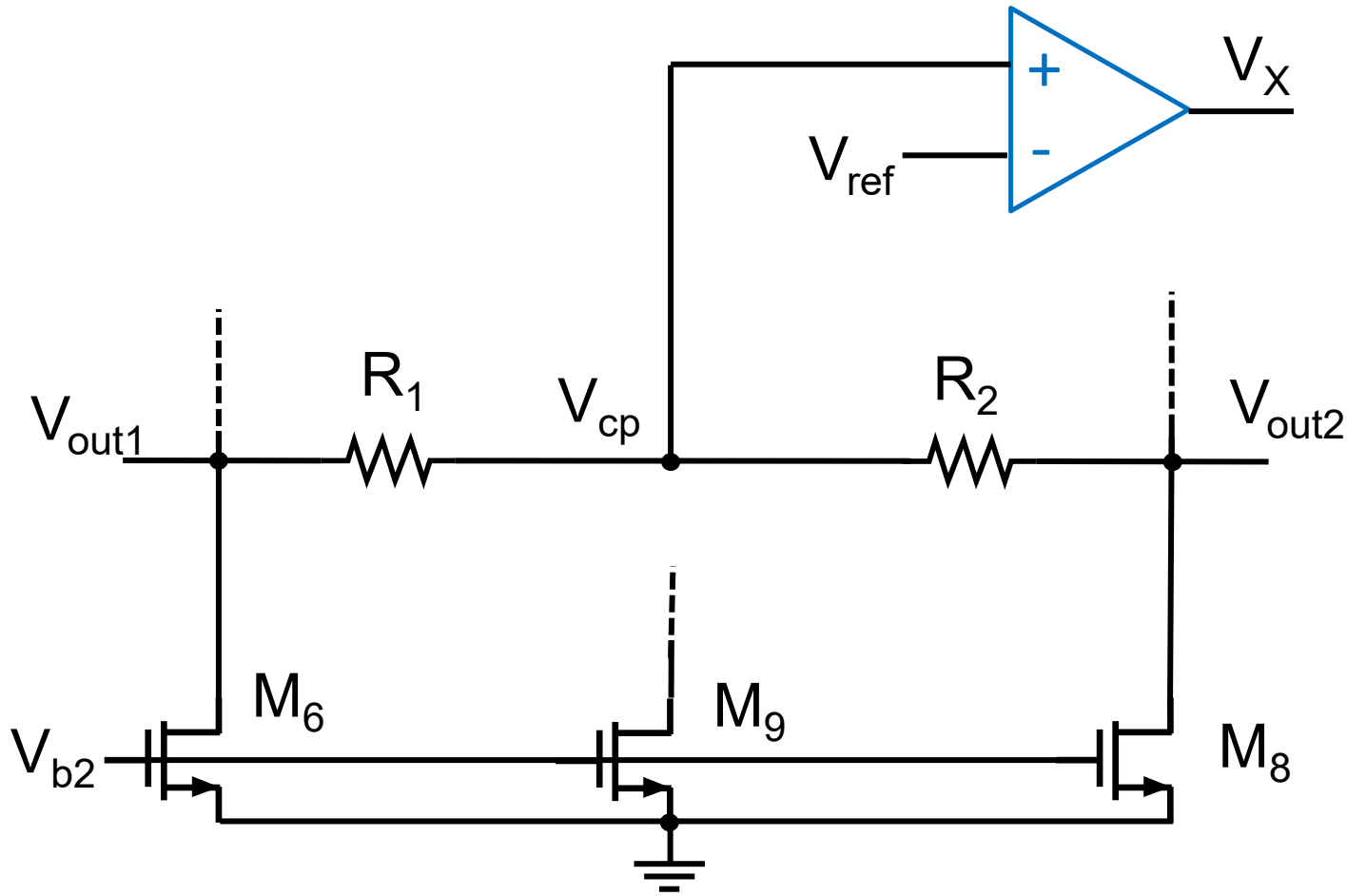
Assumption: V_{DSpsat} & $V_{DSnsat} = 0$ V



$$\text{Output Common-Mode Range (OCMR)} = V_{dd} - V_{ss} - V_{DSpsat} - V_{DSnsat}$$

The common-mode voltage V_{cm} needs to be regulated!

OP AMP DESIGN — COMMON-MODE FEEDBACK → CMFB



The operational amplifier regulates the common-mode voltages of V_{out1} and V_{out2} such that the common-mode voltage of these nodes is $V_{dd} / 2$ for maximum voltage swing.

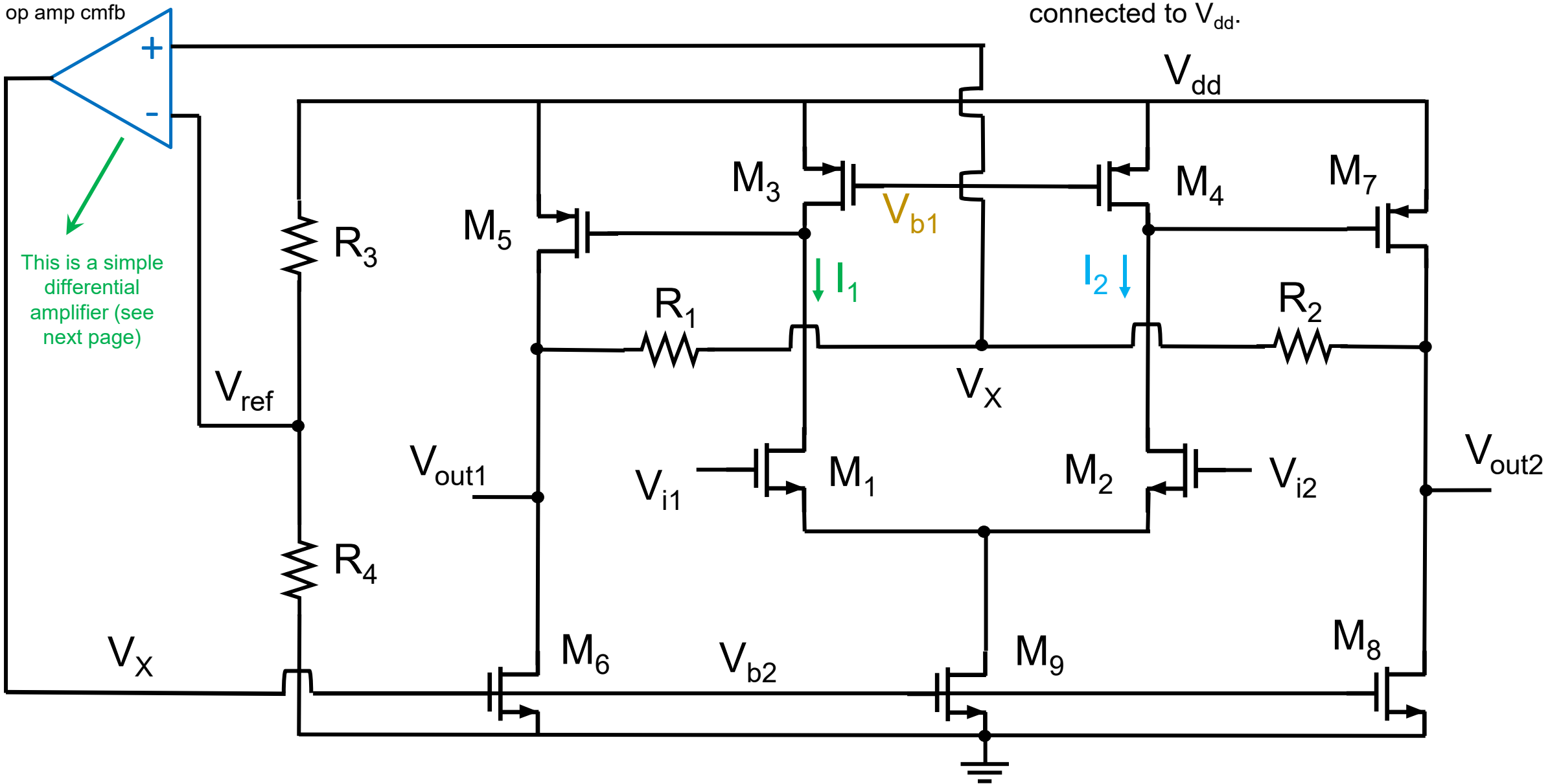
The center point of the resistors R_1 and R_2 (V_{cp}) has a voltage of

$$V_{cp} = (V_{out1} + V_{out2}) / 2$$

V_{ref} is usually generated by using a voltage divider (e.g. $10k // 10k$) which is supplied by V_{dd} . V_{ref} is then equal to $V_{dd}/2$.

OP AMP DESIGN – COMMON-MODE FEEDBACK

V_{b1} can be generated by a current mirror + bandgap circuit or a voltage divider connected to V_{dd} .



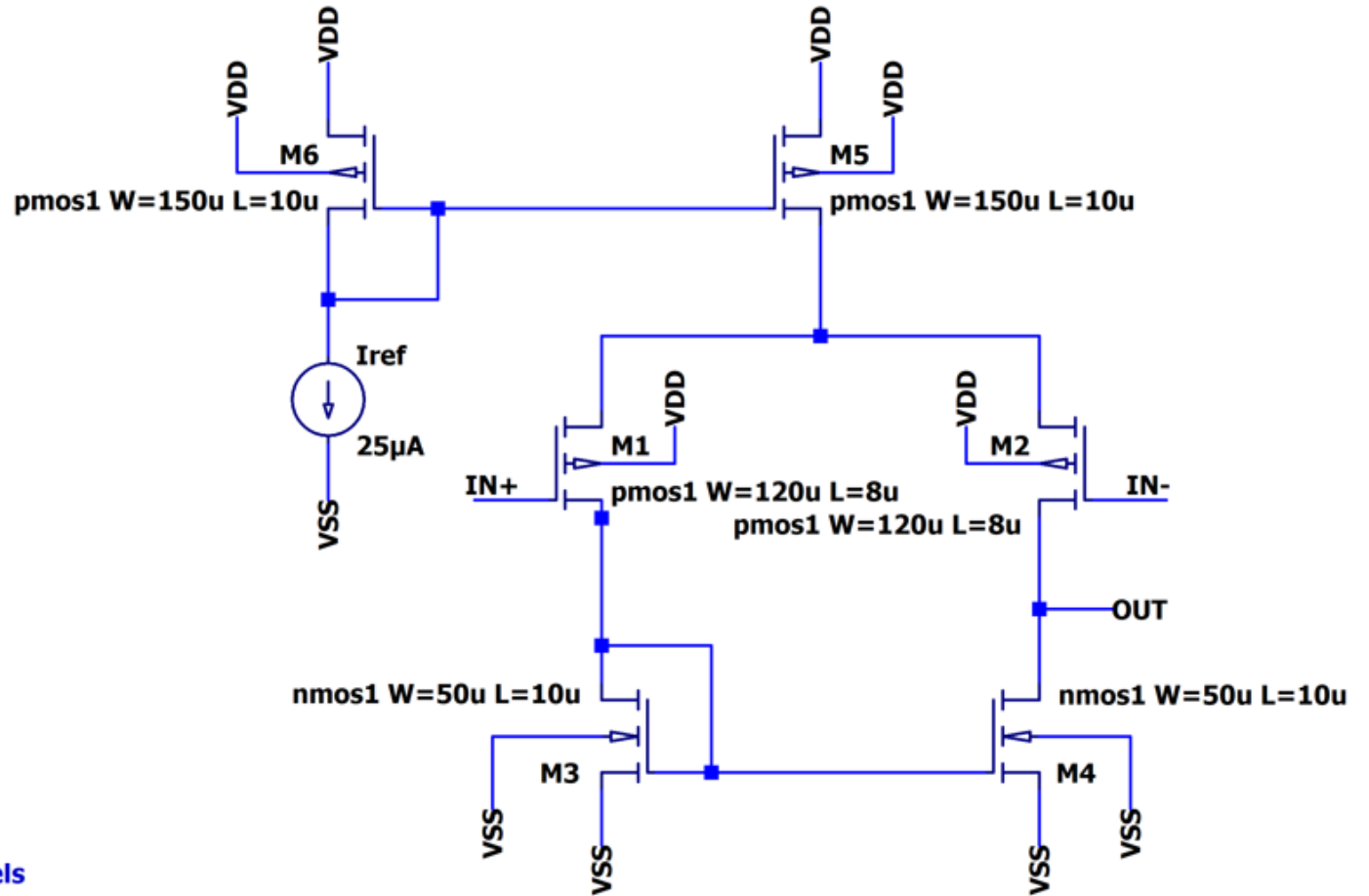
OP AMP DESIGN — COMMON-MODE FEEDBACK

If V_X is larger than $V_{DD} / 2$ (or in other words V_{ref}), then the output voltage of op amp cmfb increases. As a result of the voltage increase at the op amp output, M6 and M8 conduct more current, since $V_{GS-6,8}$ increases. Therefore, V_{out1} and V_{out2} get smaller until they become equal to $V_{DD} / 2$.

If V_X is smaller than $V_{DD} / 2$ (or in other words V_{ref}), then the output voltage of op amp cmfb decreases. As a result of the voltage decrease at the op amp output, M6 and M8 conduct less current, since $V_{GS-6,8}$ decreases. Therefore, V_{out1} and V_{out2} get larger until they become equal to $V_{DD} / 2$.

OP AMP DESIGN – COMMON-MODE FEEDBACK

Simple Differential Amplifier for Common-Mode Feedback $\rightarrow$ gm stage consists of pmos transistors



High-Swing Cascode Current Mirror

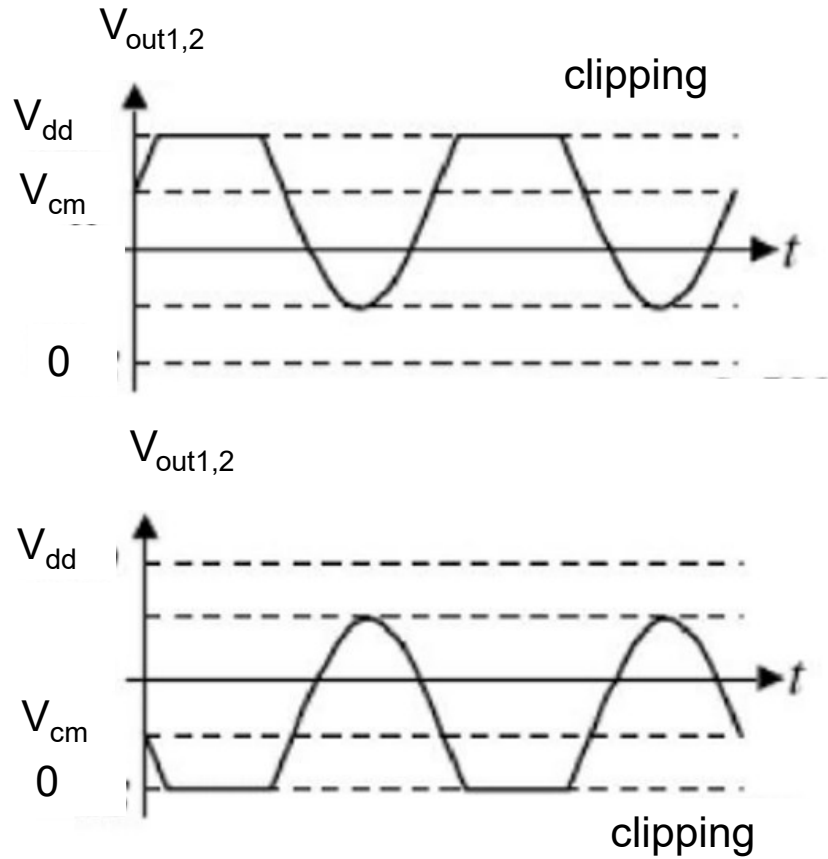
Device Models

```
.model nmos1 nmos (kp=10u Vto=+1V lambda=0.04 gamma=0.9 )
```

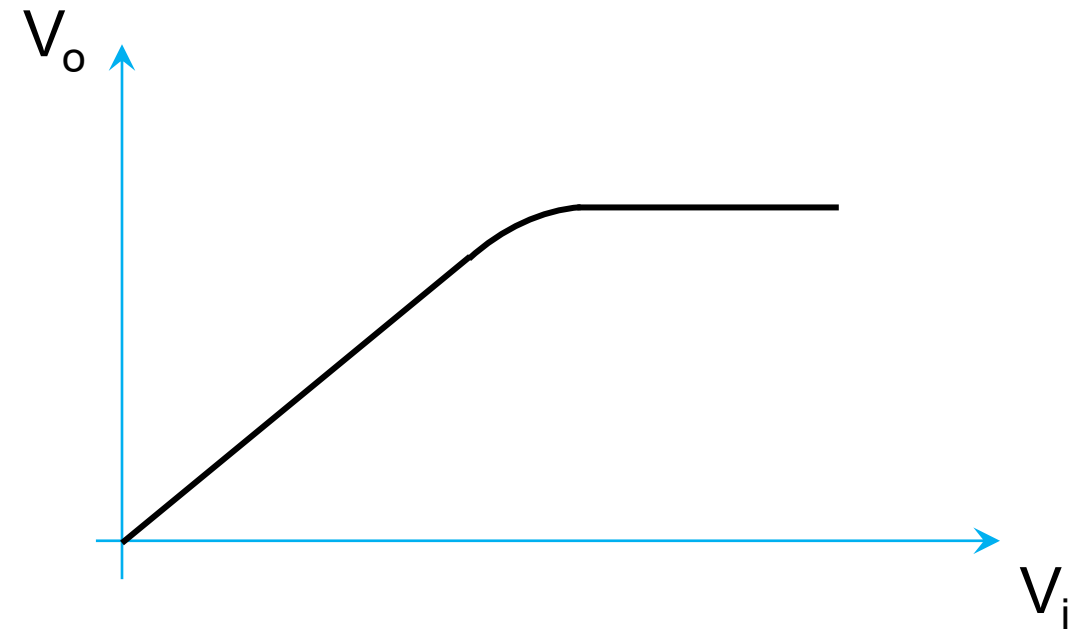
```
.model pmos1 pmos (kp=10u Vto=-1V lambda=0.04 gamma=0.9 )
```

If $I_1 \neq I_2$ due to mismatch between the two differential paths, how will the CMFB be affected?

OP AMP DESIGN — CLIPPING AND NON-LINEARITY



The voltage clipping leads to a non-linearity at higher input voltages!



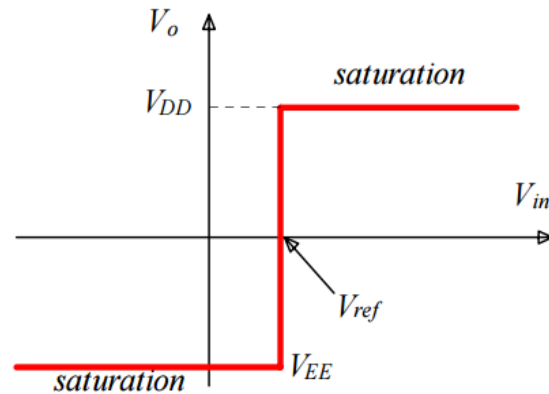
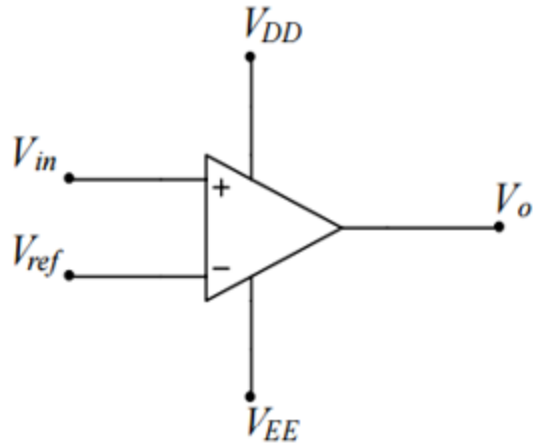
OP AMP DESIGN — COMPARATOR

Why do we need a comparator:

- As the name implies, a comparator compares the signals at its two inputs. If the signal amplitude at its (+) input is larger than the signal amplitude at its (-) input, the output is Logic High vice versa.
- Some example applications of a comparator
 - Detection of a overtemperature condition by means of a reference voltage (temperature threshold) and the temperature sensor output
 - Detection of a overvoltage condition by means of a reference voltage (temperature threshold) and the measured voltage
 - Detection of a overcurrent condition by means of a reference voltage (temperature threshold) and the current sensor output

OP AMP DESIGN — COMPARATOR

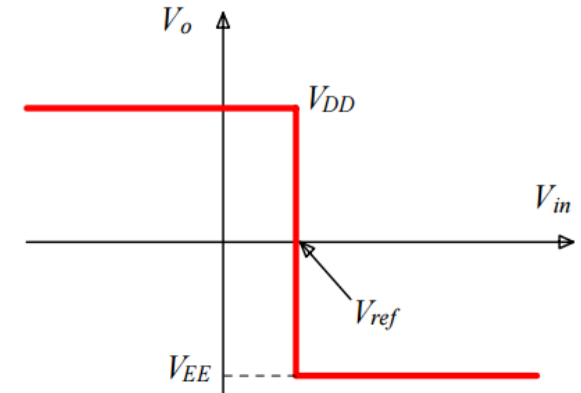
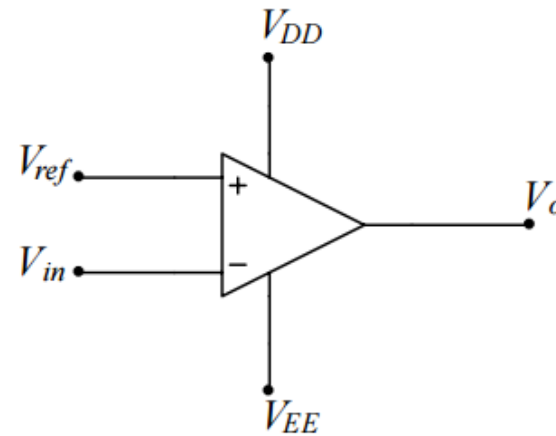
The basic comparator circuit is an op-amp arranged in the open-loop configuration as shown in the circuit of the figure below. The op-amp is characterized by an open-loop gain A and let's assume that the output voltage V_o can go all the way to V_{DD} and V_{EE} . The output voltage is given by $V_o = A (V_+ - V_-)$



Non-inverting comparator

Transfer plot of an ideal non-inverting comparator without hysteresis

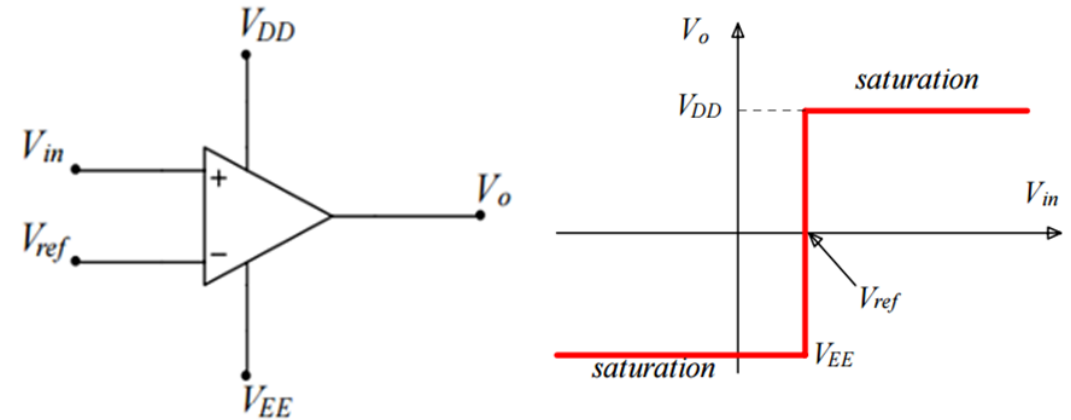
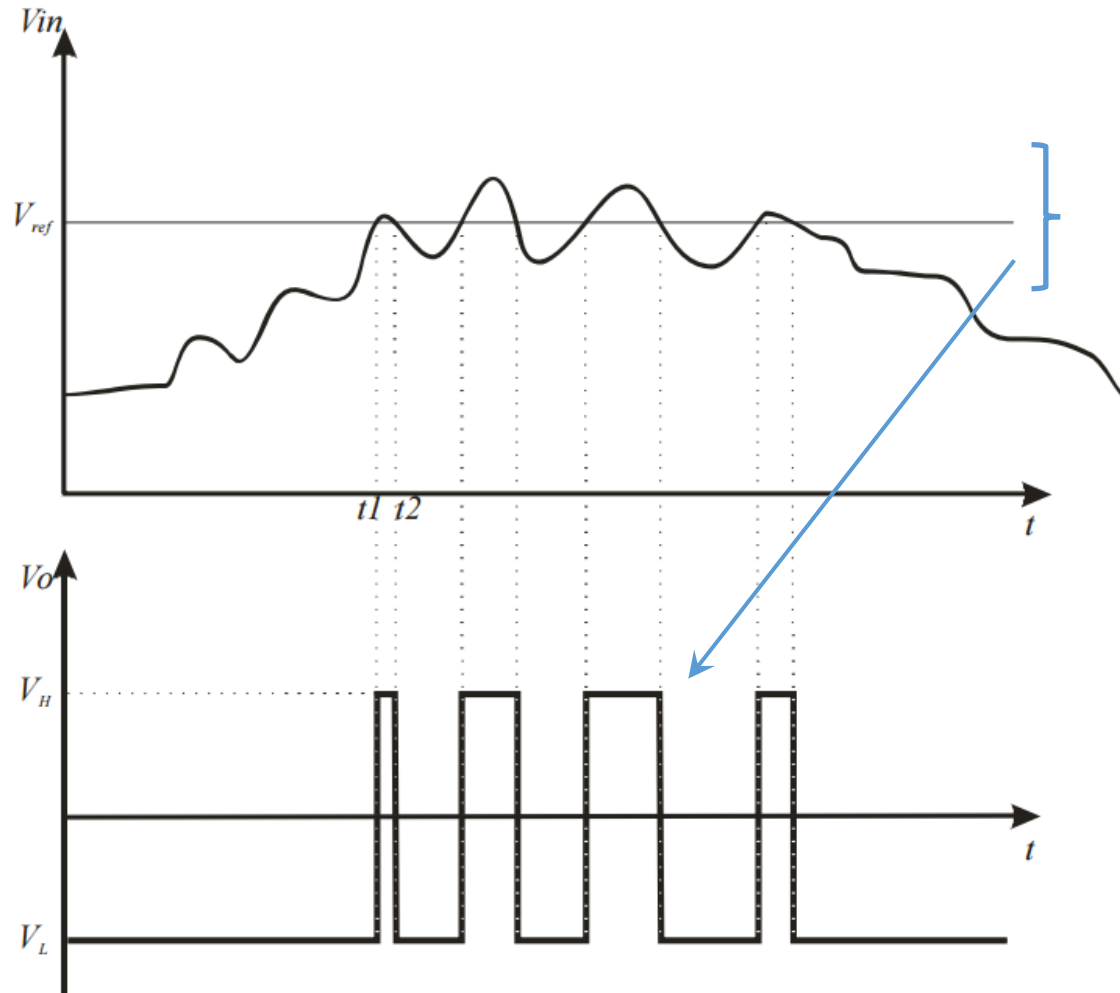
non-inverting comparator



Inverting comparator

inverting comparator

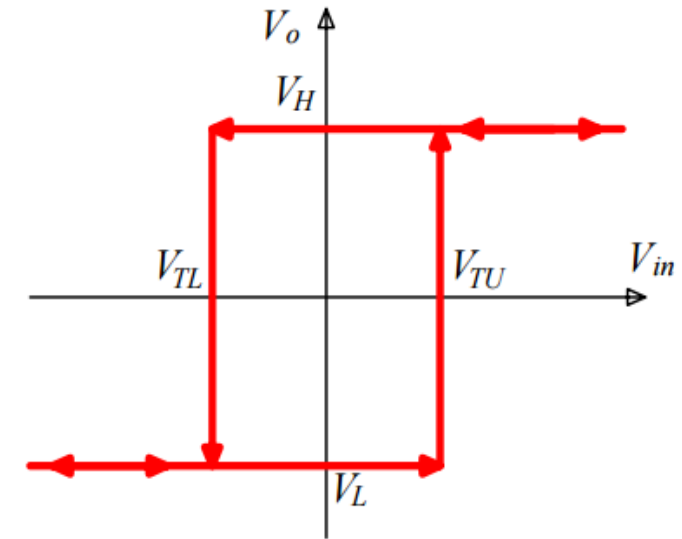
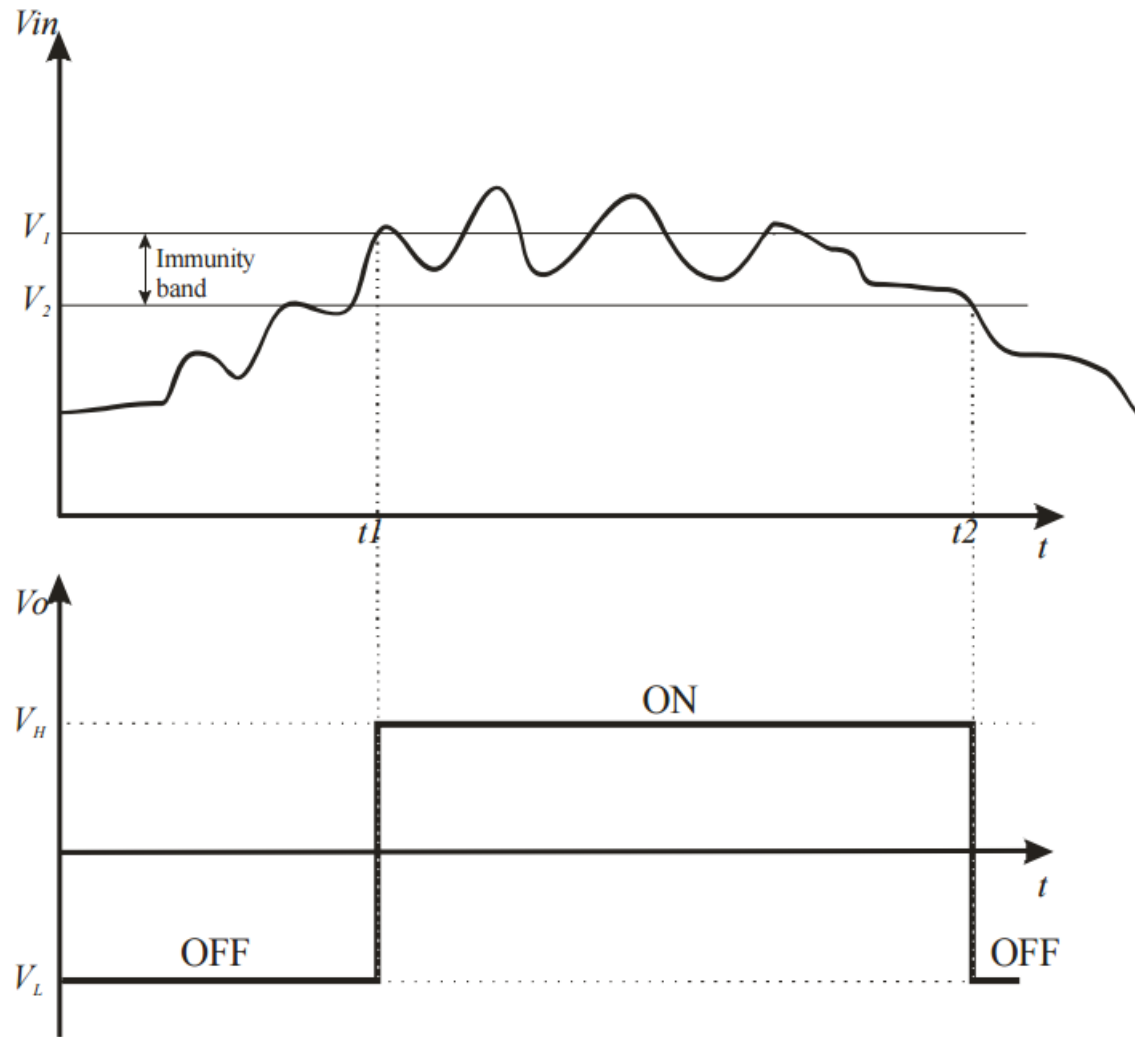
OP AMP DESIGN — COMPARATOR



If there is some noise in V_{in} , this leads to a false activation at the comparator output!

There is no noise immunity!

OP AMP DESIGN — COMPARATOR



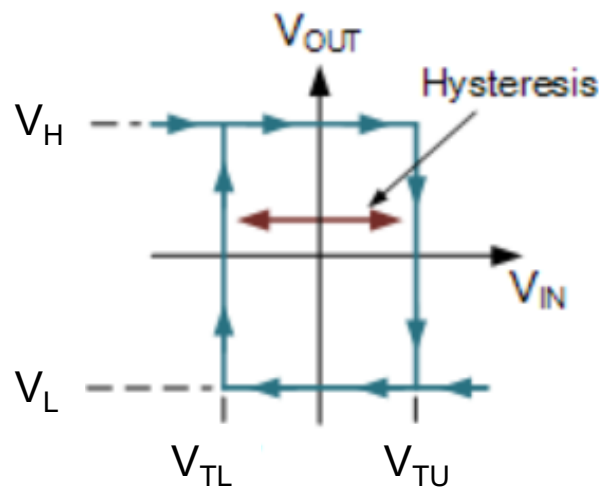
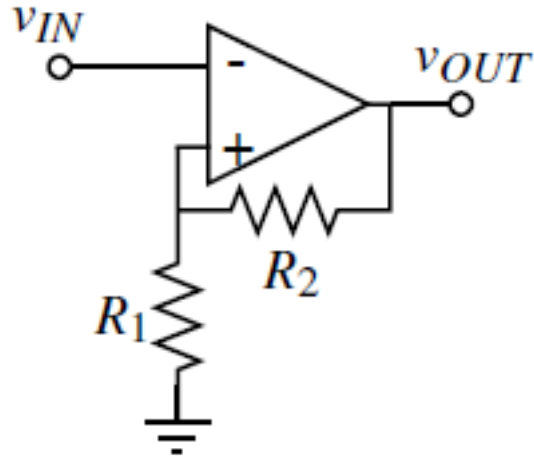
With the introduction of **hysteresis**, noise immunity is achieved

Hysteresis is achieved by the use of positive feedback

- Externally
- Internally

OP AMP DESIGN — COMPARATOR, EXTERNAL FEEDBACK

Inverting Comparator:



In the inverting comparator circuit, the input voltage V_{IN} is connected to the inverting terminal of the operational amplifier. A voltage divider network, formed by resistors R_1 and R_2 , supplies positive feedback by directing a portion of the output voltage to the non-inverting terminal. The feedback level provided to the non-inverting input is determined by the ratio of the resistances R_1 and R_2 , which is expressed as:

$$\beta = \frac{R_1}{R_1 + R_2}$$

When the input signal is less than the reference voltage, $V_{IN} < V_{REF}$, the output voltage will be HIGH, V_H and equal to the positive saturation voltage. As the output is HIGH and positive, the value of the reference voltage on the non-inverting input will be approximately equal to: $+\beta \cdot V$ called the Upper Trip Point or V_{TU} .

OP AMP DESIGN — COMPARATOR, EXTERNAL FEEDBACK

Inverting Comparator:

As the input signal, V_{IN} increases it becomes equal to this upper trip point voltage, V_{TU} level at the non-inverting input. This causes the comparators output to change state becoming LOW, V_L and equal to the negative saturation voltage as before.

But the difference this time is that a second trip point voltage value is created because a negative voltage now appears at the non-inverting input which is equal to: $-\beta \cdot V$ as a result of the negative saturation voltage at the output. Then the input signal must now fall below this second voltage level, called the Lower Trip Point or V_{TL} for the voltage comparators output to change or switch back to its original positive state.

So we can see that when the output changes state, the reference voltage at the non-inverting input also changes creating two different reference voltage values and two different switching points. One point being called the *Upper Trip Point* (V_{TU}), while the other is called the *Lower Trip Point* (V_{TL}). The difference between these two trip points is known as **Hysteresis**.

The amount of hysteresis is determined by the feedback fraction, β of the output voltage fed back to the non-inverting input. The advantage of positive feedback is that the resulting comparator Schmitt trigger circuit is immune to erratic triggering caused by noise or slowly changing input signals within the hysteresis band producing a cleaner output signal as the op-amp comparators output is only triggered once.

So for positive output voltages, $V_{REF} = +\beta \cdot V_{DD}$, but for negative output voltages, $V_{REF} = -\beta \cdot V_{DD}$. Then we can say that the amount of voltage hysteresis will be given as:

OP AMP DESIGN — COMPARATOR, EXTERNAL FEEDBACK

Inverting Comparator:

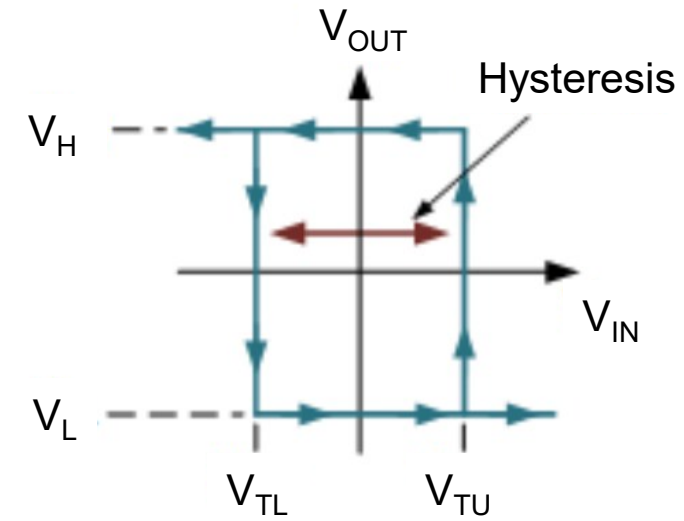
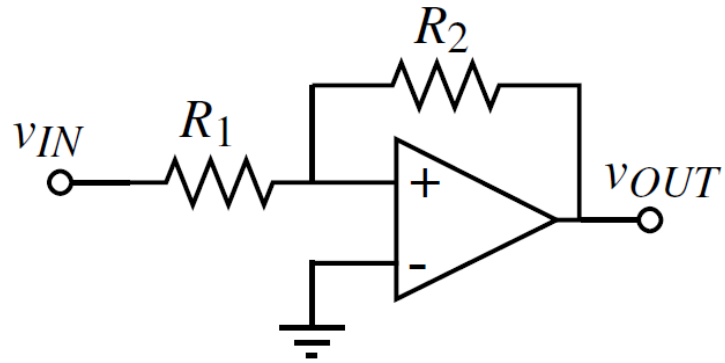
$$V_{hysteresis} = V_{TU} - V_{TL}$$

$$V_{hysteresis} = \beta V_{DD} - (-\beta V_{DD})$$

$$V_{hysteresis} = 2\beta V_{DD}$$

OP AMP DESIGN — COMPARATOR, EXTERNAL FEEDBACK

Non-Inverting Comparator:



$$V_{hysteresis} = 2\beta V_{DD} \longrightarrow \beta = \frac{R_1}{R_2}$$

References

1) www.halbleiter.org

The Principle of Semiconductor, Tokyo Electron Ltd.

2) Paul Gray, Robert Meyer, "Analysis and Design of Analog Integrated Circuits", John Wiley and Sons

3) Phillip E. Allen, Douglas R. Holberg, "CMOS Analog Circuit Design", Oxford University Press, 2002.